

Problem 1 (6.11 (c)). Use an area interpretation to evaluate the integral

$$\int_U \sqrt{1 - x^2 - y^2} dA$$

where U is the unit disk centered at the origin.

Problem 2 (6.24). Compute

$$\int_D f(x, y) dA$$

where D is the set in the plane bounded by the graphs of $y = 1$, $x = y$, and $x = 4 - y$, and $f(x, y) = e^{x+y}$.

Problem 3 (6.29). Evaluate

$$\int_R \sin y dA$$

where R is the region $0 \leq x \leq 1$, $0 \leq y \leq \sqrt{x}$.

Problem 4 (6.36). Use the change of variables $(x, y) = (u \cos v, u \sin v)$ to evaluate the integrals, where U is the unit disk centered at the origin.

(a) $\int_U \sqrt{x^2 + y^2} dA$

(b) $\int_U \sqrt{3 + x^4 + 2x^2y^2 + y^4} dA$

(c) $\int_U y dA$